CERTIFICATE

OF COMPLETION

This is to certify that

Rahma Mohamed Abdo Mohamed Elsharkawy

passed QCourse110-1 with grade 290 out of 300

Prof. Dr. Ahmed Younes
Professor of Quantum Computing
Alamein International University

Prof. Dr. Hesham M. Gaber
Vice President for Academic Affairs
Alamein International University

This summer training program was jointly organized by QWorld, Alamein International University (AIU), QEgypt and Alexandria Quantum Computing Group (AleQCG).

The QCourse110-1 was conducted fully online for undergraduate students from Egypt and Arab-speaking countries, spanning from 30 June 2025 to 15 September 2025. The rigorous 12-week curriculum included 20 hours of lectures and 30 hours of hands-on programming lab work, supplemented by a dedicated two-week period for self-study and a final project.

PROGRAM STRUCTURE

- Lecture Phase (10 weeks, 2 hours/week): This phase covers core theoretical concepts, including: Introduction to Quantum Computing & Essential Linear Algebra. Quantum Measurement, Entanglement, and Superposition. Single Qubit Gates, Controlled Gates, and Quantum Circuit Design. Quantum Protocols: Teleportation and Superdense Coding. Quantum Algorithms: Parallelism, the Deutsch–Jozsa Algorithm, and Grover's Algorithm.
- Lab Phase (10 weeks, 3 hours/week): Students gain hands-on experience through curated modules from the QWorld learning tracks (Bronze, Silver, and Nickel), applying theory to practical coding exercises.
- Project Phase (2 weeks): The program culminates in individual term projects where participants design and implement solutions based on their learning.

The total grade is out of 300 marks, comprised of:

- Quizzes & Assignments (200 marks): Based on work from the Lecture and Lab phases.
- Final Project (100 marks): Based on the individual term project.

QCourse110-1-62

Mentors: Menna Elmasry, Ahmed Saad Elfiky, Ziad Tarek Mohammed, Moataz Mohamed Fayek